1. Purpose
A VirtualBox lab for practising Active Directory administration, DNS, DHCP, shared-folder permissions, Group Policy, troubleshooting and helpdesk ticket handling.

2.
Startup Order -
- Start DC01 (Windows Server GUI) and wait for Windows to finish booting
- Start ubuntu01 and wait for its login prompt
- Start client01 and sign in with a LAB domain account
- Open http://10.77.77.30:8080 on client01 to check GLPI

3.
Shutdown Order -
- Save work and shut down client01 through Windows
- Shut down Ubuntu using 'sudo shutdown -h now'
- Shut down DC01 through Windows.
- Confirm each VM shows Powered Off in VirtualBox

Always try to use normal guest shutdowns, Power Off in VirtualBox is a last resort for if a VM is completely unresponsive.
4. Health checks

Check	Where and how	Expected result
Domain sign-in	Sign into client01 as a lab domain user; run whoami	Returns lab\username
Client network settings	On client01, run ipconfig /all	Address in 10.77.77.100–199; DNS and DHCP server 10.77.77.10
DNS lookup	On client01, run nslookup DC01.lab.test	Resolves to 10.77.77.10
Domain trust	In elevated PowerShell on client01, run Test-ComputerSecureChannel	Returns True
Shared-folder permissions	As Alice, open \\DC01\Sales and create, edit and delete a test file	All actions succeed
GLPI	On client01, open http://10.77.77.30:8080	Login page loads and sign-in works
Containers	On Ubuntu, run the commands below cd ~/homelab/glpi sudo docker compose ps	Database healthy; GLPI running

5. Troubleshooting and recovery

Symptom	Checks	Recovery
Domain account locked out	Check the account in Active Directory Users and Computers; review Security event **4740** on DC01	Wait for the configured **10-minute** lockout to expire, or unlock the account as an administrator
Account disabled	Check whether the account is disabled in Active Directory	Enable it if the user should have access, then retry sign-in
Sales share access denied	Check membership of GG_Sales, its nesting in DL_Sales_Modify, and share/NTFS permissions	Correct the identified problem; sign out and back in after group membership changes
Client cannot obtain an IP	Check VirtualBox network attachment, DHCP Server service on DC01 and client	Restore the service/network configuration, then renew the

Symptom	Checks	Recovery
address	IP settings	client’s DHCP lease
Domain names fail to resolve	Check client DNS settings and try nslookup DC01.lab.test 10.77.77.10	Restore the client’s DNS server to 10.77.77.10
Domain trust check returns False	Sign in with a local administrator and check DNS, connectivity and time	Repair the secure channel using the command below
GLPI unavailable	Check Ubuntu connectivity, container status and database/application logs	Use the logs to identify the cause; preserve database volumes during investigation

Domain Trust Repair commands -

```
Test-ComputerSecureChannel -Repair -Server DC01.lab.test -Credential (Get-Credential)
Test-ComputerSecureChannel
```

GLPI Diagnostic commands (Ubuntu) -

```
cd ~/homelab/glpi
sudo docker compose ps
sudo docker compose logs --tail=60 db
sudo docker compose logs --tail=60 glpi
```